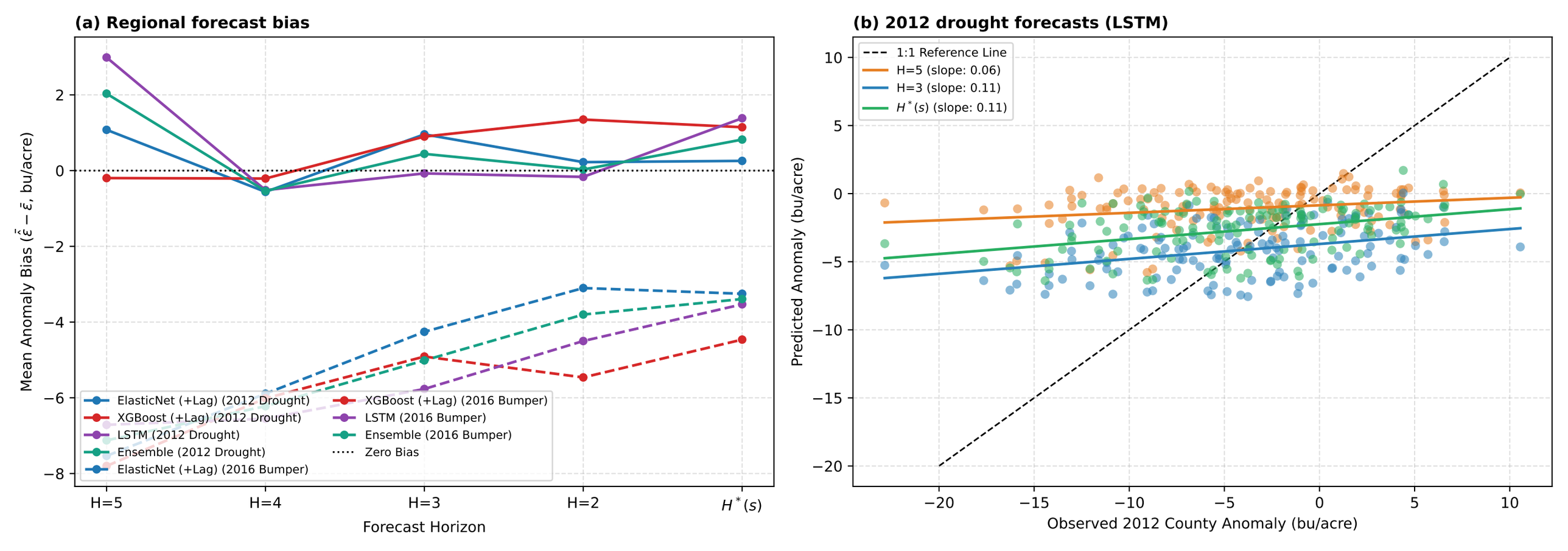
(a) Regional forecast bias
Mean Anomaly Bias ($\bar{\hat{\varepsilon}} - \bar{\varepsilon}$, bu/acre)
2
0
−2
−4
−6
−8
H=5
H=4
H=3
H=2
$H^*(s)$
Forecast Horizon
ElasticNet (+Lag) (2012 Drought)
XGBoost (+Lag) (2012 Drought)
LSTM (2012 Drought)
Ensemble (2012 Drought)
ElasticNet (+Lag) (2016 Bumper)
XGBoost (+Lag) (2016 Bumper)
LSTM (2016 Bumper)
Ensemble (2016 Bumper)
Zero Bias
(b) 2012 drought forecasts (LSTM)
Predicted Anomaly (bu/acre)
10
5
0
−5
−10
−15
−20
Observed 2012 County Anomaly (bu/acre)
1:1 Reference Line
H=5 (slope: 0.06)
H=3 (slope: 0.11)
$H^*(s)$ (slope: 0.11)